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MK : Sistem Paralel & Terdistribusi A

Dokumentasi & Analisis

Lock Manager

POST /acquire_lock
{
"resource": "file123",
"client_id": "client-5000",
"mode": "exclusive"
}
	Response 200:
	{
	"status": "lock_acquired"
	}
	
	Response 200 (shared lock added):
	{
	"status": "lock_acquired_shared"
	}
	
	Response 403 (not leader):
	{
	"error": "Not a leader"
	}
	
	Response 409 (lock conflict):
	{
	"error": "resource_locked"
	}

POST /release_lock
{
"resource": "file123",
"client_id": "client-5000"
}
	Response 200:
	{
	"status": "lock_released"
	}
	
	Response 403 (not leader):
	{
	"error": "Not a leader"
	}
	
	Response 404 (no lock found):
	{
	"error": "no_lock_found"
	}

Cache Node

POST /get
{
  "key": "file123"
}
	Response 200 (cache hit):
	{
	  "key": "file123",
	  "value": "Hello World",
	  "state": "M/S",
	  "metrics": {"hits": 1, "misses": 0, "evictions": 0}
	}
	
	Response 200 (cache miss):
	{
	  "key": "file123",
	  "value": null,
	  "state": "I",
	  "metrics": {"hits": 0, "misses": 1, "evictions": 0}
	}

POST /set
{
  "key": "file123",
  "value": "Hello World"
}
	Response 200:
	{
	  "status": "ok",
	  "key": "file123",
	  "value": "Hello World"
	}
	
	Response 400 (not leader):
	{
	  "error": "Not leader",
	  "leader_id": "node-7000"
	}

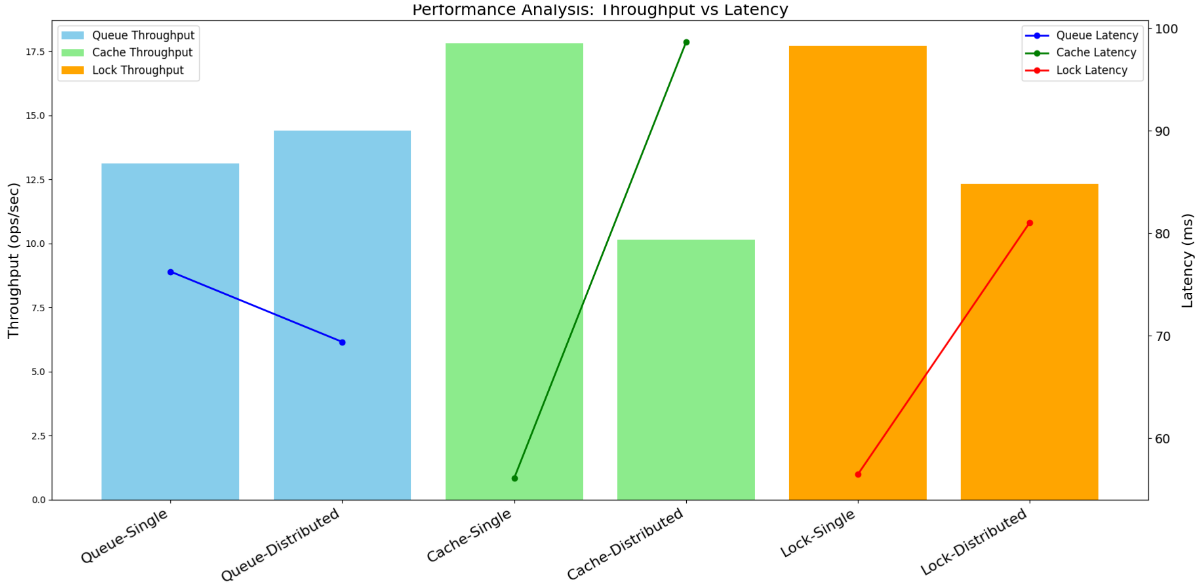
POST /invalidate
{
  "key": "file123",
  "source": "peer" # optional, default peer
}
	Response 200:
	{
	  "status": "invalidated",
	  "key": "file123"
	}

Queue Node

POST /enqueue
{
  "queue": "tasks",
  "message": "Process data",
  "client_id": "client-5000" # optional
}
	Response 200:
	{
	  "success": true,
	  "index": 5,
	  "acks": ["node-7000", "node-7001"],
	  "message": "Message enqueued to tasks via leader"
	}
	
	Response 400 (not leader):
	{
	  "error": "Not leader",
	  "leader_id": "node-7000"
	}

POST /dequeue
{
  "queue": "tasks",
  "client_id": "client-5000" # optional
}
	Response 200 (message available):
	{
	  "message": "Process data"
	}
	
	Response 200 (queue empty):
	{
	  "message": null
	}

Hasil Performa masing-masing Node



Node	Jumlah	Data	Nilai
Queue	Single	Avg Latency	76.77 ms
		Throughput	13.11 ops/sec
	Distributed	Avg Latency	69.40 ms
		Throughput	14.41 ops/sec
Cache	Single	Avg Latency	56.14 ms
		Throughput	17.81 ops/sec
	Distributed	Avg Latency	98.66 ms
		Throughput	10.14 ops/sec
Lock Manager	Single	Avg Latency	56.50 ms
		Throughput	17.70 ops/sec
	Distributed	Avg Latency	81.06 ms
		Throughput	12.34 ops/sec

Pada queue, tidak ada bottleneck yang terasa. Dari hasilnya juga sudah cukup terlihat bahwa distributed lebih unggul.

Pada cache, ada sedikit kejanggalan, yaitu single cache node jauh lebih unggul dibandingkan distributed cache node. Hal ini kemungkinan besar terjadi karena adanya overhead komunikasi karena node leader perlu melakukan sinkronisasi antar node

Pada lock manager, hasil single node juga menunjukkan hasil yang jauh lebih cepat, baik dari segi latency maupun throughput. Hal ini kemungkinan besar karena sistem pada sistem terdistribusi perlu melakukan *raft consensus* yang menyebabkan sedikit *overhead*.

Kesimpulannya, sistem terdistribusi tidak selalu lebih cepat, keuntungannya ada ketika workload dari sistem benar benar bisa dibagi secara paralel tanpa memerlukan banyak komunikasi yang bisa menyebabkan *overhead*.